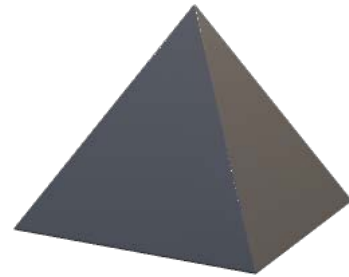
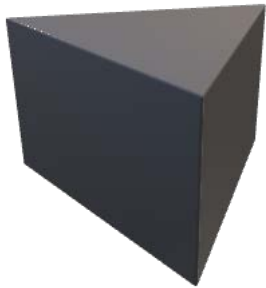


Geometry
Unit 9
Lesson 5 Practice

Name_____ **Answer Key** ____
Date_____
Period_____

Using the given three figures, answer questions 1-3.



1. Which two figures can have the same volume?
The cylinder and the prism.
2. Explain how the figures listed in Question 1 can have the same volume.
They can have congruent parallel cross-sections, but the pyramid cannot.
3. What information is required for each figure to determine if the volumes are equal?
Height and base dimensions

4. Miriam and Lloyd went to the movie theater. They each bought a container of popcorn. Miriam's container came with a square base. Lloyd's container had a circular base. Is it possible that they both purchased the same size of popcorn?



- Yes**
5. Lloyd's popcorn container was 11.5 inches high with a radius of 2.25 inches. What is the volume?
 $11.5 \cdot \pi(2.25)^2 \approx 182.8 \text{ in}^3$

6. Miriam's popcorn container was also 11.5 inches high but with a base whose side length was 2.25 inches. Determine if the two containers have the same volume.
 $2.25(2.25)(11.5) = 58.2 \text{ in}^3$
Not the same volume
7. Almond milk is sold in rectangular boxes, while evaporated milk is sold in cans. Is it possible for a given quantity of almond milk and evaporated milk to have the same volume?
Yes, if the cross-section areas are the same and the height is the same.
8. A can of evaporated milk is 4 inches tall, with a diameter of 3 inches. In a grocery store display, there are a total of 24 cans. What is the volume of milk in the display?
 $4(1.5)^2(\pi)(24) = 678.24 \text{ in}^3$
9. One box of almond milk is 4 inches tall, with a width of 2 inches and a length of 3 inches. There are 24 boxes in the display. What is the total volume of milk in the display?
 $4 \times 2 \times 3 \times 24 = 576 \text{ in}^3$
10. Compare the answers from Questions 8 and 9.
There is a larger volume of evaporated milk.